Visual Scheduler Project Progress Report

> □□□□□: **Visual Scheduler** (repository □□ □□ □□□□: *LLM Sched Copilot*)
> □□: **LLM-Assisted xv6 Scheduling Algorithm Lab** (Direction B — LLM for OS)
> LLM □□□: **Upstage Solar Pro 3** (`tools/solar_client.py`)
> □□□ □□□: **2026-05-28** (branch: `feat/upstage-runtime-strict` , base: `main`)
> □ □□□□□ □□/git history□ □□ □□□ □□□ □□□□. □□·□□ □□.

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1. Executive Summary

□ □□□□□ **xv6 (RISC-V)** □ CPU scheduler□ LLM□ “□□·□□·□□”□□ **LLM-for-OS** □□ □□□□□. LLM(Upstage Solar Pro 3)□ □□□□ □□(JSON)□ □□ **Scheduling Algorithm** □ □□□□(`tools/llm_advisor.py`), `tools/algorithm_guard.py` □ □□□□·□□□□·□□□ □□□□ □□□ □(`guard_decision.json`), `scripts/orchestrator.py` □ □□□ □□ □□ □□□ □ □□ □□□(① □□ xv6 + QEMU□ `schedtest` , ② □□□ □ Python `scheduler_simulator.py`) □ □□□ □□□□. □□ □ □□ □□ `[SCHED]` / `[SCHEDTEST]` trace□ `tools/trace_parser.py` □ □□□□□, `tools/metrics.py` □ □□ □□(response time), □□ □□(waiting time), □□ □□ (turnaround time), throughput, starvation, regret_score, judgment(SUCCESS / NEAR-SUCCESS / FAIL)□ □□□□, `tools/trace_explainer.py` □ □□ Solar Pro 3 □ □□□ □□(`trace_explanation.json`)□ □□□. `dashboard_live` (React/Vite, primary)□ `dashboard_test` (UI lab)□ □ JSON/JSONL □□ □□□□□.

□□ □□: end-to-end pipeline□ **□□□□□ □□□ □□ □□ □□**, **xv6+QEMU □□□□ `schedtest` + orchestrator□ □□ □□**(RR/FCFS/PRIORITY/MLFQ/SJF/SRTF 6□ □□□□□ `kernel/proc.c` □ `user/schedtest.c` □ □□ □□). 4□ curated profile(interactive / cpu_bound / mixed / priority_sensitive)□ □□ **xv6 snapshot 4□□□** `dashboard_live/public/live-data/snapshots/` □ □□□□ □□ dashboard□ snapshot selector□ □□ □□. GitHub Actions CI(QEMU □□ lightweight smoke + `--preview` validator + `--snapshots` □□)□ □□□ □□. **closed-loop runtime correction**□ `event_detector → correction_proposer → correction_guard` □ **preview-only □□□□□ □□□□** □□□, LLM □□ → xv6 apply → `CORRECTION_APPLIED` trace event □ **Future Work**.

□ 1.1 — Overall Progress Snapshot

Area	Status	Evidence	Notes
Project direction / README	DONE	`README.md` (29 KB, □□ □□ 2026-05-27), `CLAUDE.md`	LLM-for-OS □□□ □□□ □□□. README□ “Partial / Future Work” □ □□□□□□ □□.
Architecture design	DONE	`architecture_diagram.md` (17 KB), `docs/architecture.md`, `docs/orchestrator_design.md`	Before/Running/After 3-phase □□, JSON/JSONL □□□□□□ □□□□ □□.
Workload definition	DONE	`workloads/*.json` 6 □	interactive_heavy / long_cpu_bound_first / mixed / priority_sensitive / short_jobs / starvation_risk.
Workload analyzer	DONE	`tools/workload_analyzer.py` (209 □□)	□□ □□ + starvation_risk heuristic.
LLM advisor	DONE	`tools/llm_advisor.py` (412 □□), `tools/solar_client.py` (228 □□)	Upstage Solar Pro 3 OpenAI-compatible API, stdlib□ □□ (□□ SDK □□).
Algorithm guard	DONE	`tools/algorithm_guard.py` (626 □□)	□□□□/□□□/□□□□ □□ + algorithm×metric compatibility matrix.
Scheduler simulator	DONE	`tools/scheduler_simulator.py` (484 □□)	RR/FCFS/PRIORITY/MLFQ/SJF/SRTF 6 □□□□ □□.
xv6 scheduler integration	DONE	`xv6-riscv/kernel/proc.c` (1172 □□), `xv6-riscv/user/schedtest.c` (195 □□), syscalls `setscheduler` / `getscheduler`	6 □□□□ □□ □□ □□. `schedtest` □ □□/□□□□□ □□□ fork.
Trace generation	DONE	`[SCHED]` / `[SCHEDTEST]` □□ → `tools/trace_parser.py` (233 □□)	ARRIVE / DISPATCH / PREEMPT / EXIT / QUEUE_CHANGE □ □□□.
Metrics evaluator	DONE	`tools/metrics.py` (924 □□)	response/turnaround/waiting/throughput/max_wait/preemption/burst_prediction_e
LLM recommendation evaluator	DONE	`tools/metrics.py` `compute_judgment()` + `pick_best_algorithm()` + `compute_regret()`	regret_score □□ SUCCESS / NEAR-SUCCESS / FAIL + starvation override.
Prompt feedback loop	PARTIAL	`tools/llm_advisor.py --mode feedback`	FAIL □□ □ `feedback_rules.md` □□ □□□ □□. □□ □□□/□□□□ □□□ □□ □□□ □.
Dashboard visualization	DONE	`dashboard_live/` (React/Vite, 17 components), `dashboard_test/` (UI lab)	live-data JSON polling + Gantt + ProcessLanes + TraceStack + 4 □ snapshot □□.
Test / demo readiness	DONE (with caveats)	`scripts/final_demo_check.py`, `scripts/multi_profile_demo_check.py`, GitHub Actions `.github/workflows/`	xv6 snapshot 4□ □□ SUCCESS (regret 0.0). Live xv6 □□□ QEMU □□.
Documentation	DONE	`docs/*.md` 27 □ (architecture, demo_runbook, demo_checklist, presenter_script, RC report, audit □□□)	□ PR □□ audit □□□ □ □□ □□ □□.
Runtime correction (closed loop)	PARTIAL	`tools/event_detector.py`, `tools/correction_proposer.py`, `tools/correction_guard.py`	preview-only. xv6 apply / `CORRECTION_APPLIED` event □ Future Work.
Feedback Rule Generator (auto)	PARTIAL	`tools/llm_advisor.py --mode feedback`	FAIL □□□ □□□□ □□.

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2. Project Goal and Scope

2.1 □□□ □□

CPU scheduling□ OS □□□ □□□□□, □□ □□□ □□ “□□ print log”□ “□□□ □□” □□□ □□□□. □ □□□□□ □□ □ □□□ □□□□.

1. **xv6 □□ □□ OS □□** □□ CPU scheduling □□□□□ □□ □□ (RR / FCFS / PRIORITY+Aging / MLFQ / SJF / SRTF).
2. **LLM (Solar Pro 3)** □ □□□□ □□□ □□□□ □□□□□ □□ → Guard □ □□ → xv6 □□ → □□□□□ □□ → LLM □ □□□ □□□□ □□.

2.2 □□ LLM □□ / Python scheduling calculator □ □□ □

□□ □□	□□□
□□ LLM □□	LLM□ “□□ □□”. □ □□□□□ LLM□ **□□□ □□ xv6 □□□ □□, metrics□ □□, GUI□ □□**. LLM□ □□ □□□ regret_score□ □□ □□□□.
Python scheduling calculator	□□□□□ □□□□□ □□□□□□ □. □ □□□□□ □□ □□□□□ **□□ xv6 schedtest**(`xv6-riscv/user/schedtest.c`) □ QEMU□□ □□ □□ [SCHED] log □ □□□□.

2.3 OS □□□ □□□□□ □□

OS □□	□□□ □□□□□□
process	`xv6-riscv/kernel/proc.c` `struct proc` (ctime, queue_level, wait_ticks, predicted_burst □ □□ □□ □□□)
process state	RUNNING / RUNNABLE / SLEEPING / ZOMBIE — kernel + simulator □□ □□
ready queue	RR / MLFQ □□ RUNNABLE pool□□ □□; MLFQ□ `queue_level 0..2` 3-queue □□
CPU scheduling	`proc.c` `scheduler()` switch by `sched_mode` (RR/FCFS/PRIORITY/MLFQ/SJF/SRTF)
preemption	RR / MLFQ / SRTF: `quantum_expired` PREEMPT event, □□ □□□□ □□ □ □□□□ yield
system calls	`setscheduler(int mode)` / `getscheduler(void)` — `xv6-riscv/kernel/sysproc.c:115,127`, `kernel/syscall.c:104-105,135-136`, `user/user.h:27-28`
metrics-based evaluation	`tools/metrics.py` — response / turnaround / waiting / throughput / starvation / preemption_count / burst_prediction_error

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3. System Architecture

3.1 □□ □□ □□ (host orchestrator □□)

```
workloads/*.json
   └─► tools/workload_analyzer.py          → outputs/.../workload_summary.json
          └─► tools/llm_advisor.py (Solar Pro 3)
                  └─► recommendation.json
                         └─► tools/algorithm_guard.py
                                 └─► guard_decision.json
                                        └─► scripts/orchestrator.py
                                              ├─ backend=xv6  : QEMU + xv6 + schedtest
                                              │                  → outputs/xv6_raw_<algo>_seed<seed>.log
                                              │                  → tools/trace_parser.py → trace_<algo>.jsonl
                                              └─ backend=simulator: tools/scheduler_simulator.py
                                                                   → trace_<algo>.jsonl
                                                                   └─► tools/metrics.py → metrics.json
                                                                         ├─► tools/event_detector.py → runtime_events.json
                                                                         │     └─► tools/correction_proposer.py → correction_proposal.jso
                                                                         │              └─► tools/correction_guard.py
                                                                         │                     → correction_guard_decision.json (previe
                                                                         ├─► tools/trace_explainer.py (Solar Pro 3) → trace_explanation.j
                                                                         └─► dashboard_live/public/live-data/ (manifest.json + □ □□ □□□)
                                                                                 → dashboard_live (React/Vite, port 5174)
```

□□□□□□ □□□ `all_metrics.csv`, `evaluation_result.csv` □□ **CSV** □□□ □ repo□□ □□□□ □□□. `CLAUDE.md` □□□ □□ **□□ □□ □□□□□□ JSON/JSONL** □□, evaluator□ `metrics.json` □□ `judgment` / `regret_score` / `best_algorithm` □□□ □□□ □□□□.

3.2 □□□ □□ / □□□ / □□

#	Stage	□□	□□	□□ □□	□□
1	Workload Definition	(□□)	`workloads/*.json`	`workloads/`	DONE — 6 □
2	Workload Analyzer	`workloads/<name>.json`	`workload_summary.json`	`tools/workload_analyzer.py`	DONE
3	LLM Advisor (Solar Pro 3)	`workload_summary.json` (+ optional `feedback_rules.md`)	`recommendation.json`	`tools/llm_advisor.py`, `tools/solar_client.py`	DONE
4	Algorithm Guard	`recommendation.json`	`guard_decision.json`	`tools/algorithm_guard.py`	DONE
5a	xv6 Execution	`guard_decision.json` + `xv6-riscv/user/schedtest`	`outputs/xv6_raw_<algo>_seed<seed>.log`	`scripts/orchestrator.py` (QEMU □□□), `kernel/proc.c`, `user/schedtest.c`	DONE (□□ + orchestrator)
5b	Simulator Execution	`guard_decision.json` + workload	`trace_<algo>.jsonl`	`tools/scheduler_simulator.py`	DONE
6	Trace Parser	raw xv6 log	`trace_<algo>.jsonl`	`tools/trace_parser.py`	DONE
7	Metrics Evaluator	`trace_<algo>.jsonl` (+ `recommendation.json`)	`metrics.json` (judgment □□)	`tools/metrics.py`	DONE
8	Event Detector	trace + metrics	`runtime_events.json`	`tools/event_detector.py`	DONE
9	Correction Proposer	`runtime_events.json` + recommendation	`correction_proposal.json` (preview only)	`tools/correction_proposer.py`	PARTIAL (preview only, xv6 apply □□)
10	Correction Guard	`correction_proposal.json`	`correction_guard_decision.json` (preview only)	`tools/correction_guard.py`	PARTIAL
11	Trace Explainer (Solar Pro 3)	trace + metrics	`trace_explanation.json`	`tools/trace_explainer.py`	DONE
12	Feedback Rule Generator	`metrics.json` (judgment=FAIL)	`feedback_rules.md`	`tools/llm_advisor.py --mode feedback`	PARTIAL (□□ □□)
13	Dashboard	live-data □□ □□	(browser)	`dashboard_live/src/App.jsx` □ 17 components	DONE
14	Snapshot exporter	live-data + manifest	`live-data/snapshots/<profile>/`	`scripts/export_profile_snapshots.py`	DONE
15	Contract validator	live-data, snapshots, preview	(exit code)	`tools/validate_dashboard_contract.py` (`--preview`, `--snapshots` □□)	DONE

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4. Implemented Features

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4.1 Workload Definition / Sample Workloads

- **Status:** DONE.
- **Purpose:** □□□□ □□□ □□□ □□□□.
- **Related files:** `workloads/interactive_heavy.json` (25□ interactive process), `long_cpu_bound_first.json`, `mixed_workload.json`, `priority_sensitive.json`, `short_jobs.json`, `starvation_risk.json`.
- **Schema:** list of `{pid, arrival_time, cpu_bursts:[int], io_bursts:[int], priority, label}`.
- **Evidence:** `workloads/interactive_heavy.json:1-292` — 25 short interactive jobs.

4.2 Workload Analyzer

- **Status:** DONE.
- **Purpose:** workload JSON → □□ summary (LLM□□ □ "□□ □□□ □□").
- **Related files:** `tools/workload_analyzer.py:39-147`.
- **Output keys:** `process_count`, `avg_arrival_gap`, `cpu_bound_ratio`, `interactive_ratio`, `avg_priority`, `priority_variance`, `has_starvation_risk` (priority_max ≥ 3 × priority_min heuristic), `burst_count_distribution`, `total_cpu_work`, `workload_file`.
- **Note:** □□ burst □□ □□□□□ "□ burst □"□ □□□□ □ — LLM□□ summary□ □□□□, □□ □□ burst□ **SJF/SRTF predictor □□□□ □□□ noise □□ □□□□□ □□□** (`CLAUDE.md` □ burst prediction rule □□).

4.3 LLM Scheduling Advisor (Upstage Solar Pro 3)

- **Status:** DONE.
- **Purpose:** workload summary → □□□□ □□ + □□□□ + □□.
- **Related files:** `tools/llm_advisor.py:50-167, 313-356` (advise mode), `tools/solar_client.py` (□□).
- **How it works:**
  1. `SolarClient` □ `.env` □ `UPSTAGE_API_KEY` □ stdlib `urllib.request` □ `POST https://api.upstage.ai/v1/chat/completions` (□□ `solar-pro3`, OpenAI-compatible □□□).
  2. system prompt□ 6□ □□□□□ params schema□ □□ (`tools/llm_advisor.py:72-103`).
  3. JSON mode (`response_format={"type":"json_object"}`) □ strict JSON □□ → `validate()` □ □□□□ enum, reason, params □□ □□.
  4. `schema_compat.normalize_algorithm_name` □□ dashboard □□ `recommended_scheduling_algorithm` □□□ □□ □□.
- **Input:** `outputs/workload_summary.json` + (optional) `outputs/feedback_rules.md` .
- **Output:** `outputs/recommendation.json` (algorithm, params, reason, target_metric, confidence, `_meta`).
- **Evidence from code:** `tools/solar_client.py:81-89` — `UPSTAGE_API_KEY` □□□ □□□ □□. □ □□□ `.env` □□ □□□□ `.gitignore` □ □□ (`.gitignore:1-3`).
- **Limitations:** □ □□□ □□ □□. orchestrator□ `--offline-fixture` □□□□□ `outputs/demo/recommendation.json` □□ □□ (□ □□ manifest□ `metadata_source=demo_fallback` □ □□□□ dashboard □ `FALLBACK` □□□ □□).

4.4 Upstage Solar Pro 3 API Integration

- **Status:** DONE (strict □□□ □□ □□ `2d67299 feat(runtime): strict Upstage Solar Pro 3 backend; opt-in demo fallback` □□ □□).
- **Related files:** `tools/solar_client.py` □□.
- **Key design choices:**
  - □□ SDK □□□ (`requirements.txt` □□□ □□) — stdlib `urllib` □ □□. □□ □□□ `pip install` □□ □□ □ □□.
  - □□□: 429/500/502/503/504 □ □□ □□ backoff (`solar_client.py:107-130`).
  - `complete_json()` □ JSON □□□ □□ □ □ `{...}` □□ □□ fallback (`solar_client.py:202-211`).

4.5 Algorithm Guard

- **Status:** DONE.
- **Purpose:** LLM □□□ “xv6□□ □□□ □□ □□□□”□ □□.
- **Related files:** `tools/algorithm_guard.py:1-120, 121-626` .
- **□□ □□:**
  1. JSON □□□ / □□□□ enum (`SUPPORTED_ALGORITHMS = [FCFS, RR, PRIORITY, MLFQ, SJF, SRTF]`).
  2. metric enum + alias □□□ (`avg_response_time` → `response_time` , …).
  3. Algorithm×Metric Compatibility Matrix (algorithm_guard.py:66-115) — □: `PRIORITY × starvation = 0.2` □ REJECT, `MLFQ × response_time = 0.95` □ APPROVE.
  4. Confidence □□□ (`CONFIDENCE_REJECT_THRESHOLD = 0.3`).
  5. □□□□ □□ (`PARAM_RANGES`): RR.quantum 1-100, MLFQ.queues 2-5, MLFQ.quantum list 1-100, aging_threshold 1-10000, boost_interval 10-10000, SJF/SRTF.alpha_percent 0-100 □.
- **Output:** `guard_decision.json` (`accepted` / `rejected` + `fallback_scheduling_algorithm`).
- **Limitations:** □ □ §1 Notes □□ SJF/SRTF□ predictor □□□ □□□ □□ □□□□□ □□ □□.

4.6 Scheduler Simulator (host-side dev/fallback)

- **Status:** DONE (□□/□□□ / “xv6 □□ □□ □□”).
- **Related files:** `tools/scheduler_simulator.py` (484 □□).
- **Implemented algorithms:** RR (`_pick_rr`), FCFS (`_pick_fcfs`), PRIORITY+aging (`_pick_priority` , 20-tick aging step), MLFQ (`_pick_mlfq` , 3 □ + aging promotion `QUEUE_CHANGE` □□□), SJF / SRTF (predictor □□).
- **Trace emission:** `Tracer.emit(tick, event, pid, ...)` → JSONL.
- **Events emitted:** ARRIVE, DISPATCH, PREEMPT, EXIT, QUEUE_CHANGE (+ SLEEP/WAKEUP □□□ io_bursts □□ □).

4.7 xv6 Kernel Scheduling Algorithms

- **Status:** DONE — 6□ □□ □□ □□.
- **Related files:** `xv6-riscv/kernel/proc.c` (1172 □□, `sched_mode` switch).
- **Key constants/functions:**
  - `sched_mode` enum 0..5 (RR, FCFS, PRIORITY, MLFQ, SJF, SRTF) — `proc.c:31` .
  - `MLFQ_BOOST_THRESHOLD = 20` □□□ (`proc.c:586`).
  - 5 picker □□: `pick_rr/pick_fcfs/pick_priority/pick_mlfq/pick_sjf/pick_srtf` (□□ `sched_debug` □ `[SCHED]` □□ emit).
  - burst predictor: `proc.c:518` `// ── Burst predictor API (SJF/SRTF) ──` □□ — exponential averaging.
  - **System calls:** `setscheduler(mode)` / `getscheduler()` — `sysproc.c:115` , `syscall.c:104,135` , `user/user.h:27` .
- **Trace lines:** `[SCHED] tick=.. algo=.. event=DISPATCH|PREEMPT|EXIT|QUEUE_CHANGE pid=.. queue=..` plus `[SCHEDTEST]` metadata.
- **schedtest workloads:** `xv6-riscv/user/schedtest.c:30-65` 4□ curated profile(interactive, cpu_bound, mixed, priority_sensitive). □□ □□□ □ □□□ □□ (random □□□ future).

4.8 Trace Event Generation (ARRIVE / DISPATCH / PREEMPT / EXIT)

- **Status:** DONE in both backends.
- **xv6 path:** `proc.c` `sched_trace_*` □□□ console□ `[SCHED]` □□ □□ → `trace_parser.py` □ □□□.
- **Simulator path:** `scheduler_simulator.py` `Tracer.emit()` □ □□ JSONL □□.

- **Schema:** `{"tick":int,"algo":str,"event":str,"pid":int, ...}` (`docs/trace_format.md` □□).
- **Field alias:** `tools/metrics.py:62-66` □ simulator□ `time` / `algorithm` □□□ □□□□.

4.9 Metrics Calculation

- **Status:** DONE.
- **Related files:** `tools/metrics.py:517-631` (`compute_metrics`).
- **Computed metrics:**

Metric	Formula / Source		
response_time	first_run_time − arrival_time		
turnaround_time	finish_time − arrival_time		
waiting_time	turnaround_time − total_cpu_burst (kernel□ EXIT □□□ □□ □□; □□□ trace dispatch □□□□ □□□)		
throughput	completed_count / total_execution_time		
max_waiting_time	per-process waiting□ max		
starvation_occurred	`waited > 3 × avg_waiting AND waited ≥ 5 ticks` (`STARVATION_MULTIPLIER=3` , `MIN_STARVATION_WAIT_TICKS=5`)		
preemption_count	trace PREEMPT □□□ □□□		
burst_prediction_error	SJF/SRTF□; \	predicted − actual\	□ □

- **Limitation (□□□□):** xv6 trace □ □□ □(EXIT 5□ □□) □□ 3× □□□□□ 1-tick □□□ starvation□□ □□ FAIL□ □□ □□□ □□ PR #14 (`fix(metrics): add absolute floor to starvation rule for short xv6 traces`) □ absolute floor 5 ticks□ □□□. □□ □□□ regret □□□□ absolute floor (#15).

4.10 Evaluator (SUCCESS / NEAR-SUCCESS / FAIL)

- **Status:** DONE.
- **Related files:** `tools/metrics.py:414-489` .
- **Judgment logic** (`compute_judgment(regret_score, starvation_occurred)`):

```
if starvation_occurred: return "FAIL"
if regret_score is None: return "UNKNOWN"
if regret_score <= 0.10: return "SUCCESS"
if regret_score <= 0.30: return "NEAR-SUCCESS"
return "FAIL"
```

- `regret_score = (llm_metric − best) / best` (lower-is-better), `(best − llm_metric) / best` (higher-is-better — throughput).
- `best_algorithm` = target_metric □□ □□ □□□□ □ □□.
- □□ baseline □ □□□ `_make_synthetic_rr_baseline()` (PR `d546c20`) □□ □□ RR baseline □□ — regret□ □□ □□.

4.11 Prompt Feedback Loop

- **Status:** PARTIAL.
- **Related files:** `tools/llm_advisor.py:170-305` (`run_feedback`), `FEEDBACK_SYSTEM_PROMPT` .
- **How it works:** `python3 tools/llm_advisor.py --mode feedback` □□ □□ □□ □ metrics.json□ judgment□ FAIL□□ Solar Pro 3□□ "□ □□□ □□□ □"□ □ □□ `outputs/feedback_rules.md` □□. advise □□ □□ □□ □ □□□□ system prompt□ append.
- **What is missing:** orchestrator□ FAIL □□ □ □□□□ feedback □□□ □□□□□ □□ (□□). dashboard□□ rule □□ □□.

4.12 Trace Explainer (Solar Pro 3)

- **Status:** DONE.
- **Related files:** `tools/trace_explainer.py` (238 □□).
- **Output schema:** `{scheduling_algorithm, detected_pattern, summary, main_reason, evidence:[], suggestion, runtime_corrections_applied}` — dashboard `LLMExplanation.jsx` □ □□□ □□.
- **Note:** `runtime_corrections_applied` □ LLM□□ □□□ □□ trace □ `CORRECTION_APPLIED` □□□□ □□□ (`trace_explainer.py:168-169`).

4.13 Runtime Correction Preview (event_detector / proposer / guard)

- **Status:** PARTIAL — preview only, xv6 apply □□.
- **Related files:**
  - `tools/event_detector.py` (148 □□) — starvation(40 tick □□) / low_throughput(<0.05) / high_preemption_rate(>0.30/tick) / high_response_time(>10) □□.
  - `tools/correction_proposer.py` (235 □□) — □□□ □(starvation→aging_strengthen, FCFS+high_response_time→algorithm_change to RR, MLFQ→quantum □□ □).
  - `tools/correction_guard.py` (172 □□) — proposer □□□ `algorithm_guard` □ □□ PARAM_RANGES □□□□ □□□.
  - dashboard □□: `dashboard_live/src/components/RuntimeCorrectionPreview.jsx` .
- **Why preview only:** □□ □□□□ `preview_only=true / applied=false` □□. xv6 □□□ `CORRECTION_APPLIED` □□□□ setscheduler □□□□ □□□□ □□ □□. README §5.2.1 / §10 □ "Future Work" □□.
- **Validation:** `tools/validate_dashboard_contract.py --preview` □ CI□□ preview □□□ schema□ □□ (`PR #59`).

4.14 Dashboard — `dashboard_live` (Primary)

- **Status:** DONE.

- **Related files:** `dashboard_live/src/App.jsx` (256 □□) + 17 components.
- **Components (`dashboard_live/src/components/`):** `Header` , `DemoGuide` , `LLMRecommendation` , `AlgorithmGuard` , `RecommendationEvidence` , `CounterfactualMetricView` , `RuntimeCorrectionPreview` , `EvaluationResult` , `LLMExplanation` , `MainGantt` , `ProcessState` , `TraceStack` , `ProcessLanes` , `WorkloadSummary` , `AlgorithmComparison` , `MetricVisualization` , `Card` , `constants.js` .
- **Data sources (polled every 1 s in live mode):** `dashboard_live/public/live-data/` □ `manifest.json` , `recommendation.json` , `guard_decision.json` , `workload_summary.json` , `metrics.json` , `trace_explanation.json` , `runtime_events.json` , `correction_proposal.json` , `correction_guard_decision.json` , 6× `trace_<algo>.jsonl` , `snapshots_manifest.json` .
- **Snapshot selector:** header□□ 4□ xv6 profile (interactive / cpu_bound / mixed / priority_sensitive) □ □□ → `live-data/snapshots/<profile>/` □□ base □ □□ (PR #37, #38).
- **Backend badge:** `XV6 TRACE` / `SIMULATOR FALLBACK` / `FALLBACK` — manifest□ `metadata_source` □□.

4.15 Dashboard — `dashboard_test` (UI Lab)

- **Status:** DONE.
- **Related files:** `dashboard_test/src/` — 16 components (live □ □□□□□ `HeroSection` , `UITestControls` □□; `RecommendationEvidence` / `CounterfactualMetricView` / `RuntimeCorrectionPreview` / `DemoGuide` □□).
- **Purpose:** □□ fixture □ UI □□□ □□.

4.16 Dashboard — `dashboard/dashboard.py` (Streamlit, legacy)

- **Status:** DONE but **deprecated** — `docs/implementation_status.md` □ “legacy fallback only”□ □□. 1112 □□. □□ □□□ React □□□ □□.

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5. Major Code Characteristics

5.1 Design highlights

1. **□□□ “□□□ JSON □□□”□ □□.** □ □□□ □□ JSON□ □□ □□ JSON□ □□. shared state □□.

→ □□ □□□ □□ □□ □□, □□ □□□□□□ □□ □□ □□ □□.

1. **LLM□ □□ scheduler□ □□□□ □□□.** `tools/llm_advisor.py` □ *recommendation*□ □□□□, `algorithm_guard.py` □ □□□ □□□ simulator/xv6□ □□. `tools/correction_proposer.py` □ `preview_only=true / applied=false` □ □□□□. xv6 □□□ LLM□ □□ □□□□ □□□ □□ □□□□ □□.

1. **Algorithm Guard□ □□□ □□□□□.** algorithm □□□, metric mismatch, confidence □□, □□□□ □□ □□ → reject + `fallback_scheduling_algorithm` . □ □□□ □□□ LLM□ □□□ “xv6 □□ □□”□ □□□□ □□□.

1. **Trace-driven metrics.** metrics.py□ “□□ □□□□□ □□□□”□ □□ □□□ LLM□ □□ □□ **trace event(ARRIVE/DISPATCH/EXIT)□□ □□□**□□. □□ trace□ □□ □□□□ □□□ metric□ □□□.

1. **Simulator vs xv6 schema □□.** `tools/schema_compat.py` (298 □□) □ `algorithm` / `scheduling_algorithm` , `time` / `tick` , `Priority` / `PRIORITY` □ □ □□□ □□ □□□ □□. dashboard □ □ □□□ □□□□ □□ □□ □□□ □ □□ □.

1. **Dashboard contract validator.** `tools/validate_dashboard_contract.py` (479 □□) □ live-data □□□ JSON/JSONL□ dashboard □ □□ □□□□ □□□□ □ □□. `--preview` □□ □□(PR #59)□ runtime correction □□□□ □□. `--snapshots` □□(PR #36)□ □□ snapshot □□□□ + manifest cross-link □□.

1. **API key □□.**
   - `.env` □□ □□ → `.gitignore:1-3` □ git□□ □□.
   - `.env.example` □□ placeholder□ (`UPSTAGE_API_KEY=your_upstage_api_key_here`).
   - `SolarClient.__init__` □ □ □□□ □□□□□ raise (`solar_client.py:81-89`).
   - PR `0b1903d Add root .gitignore to prevent accidental credential commits` □ □□□ □□.
   - `requirements.txt` □□□ `anthropic` / `openai` SDK□ □□.
   - **□ □□□□ API □□ □□ □□□□ □□□.**

1. **Error handling.** Solar API □□□ stdlib `urllib` + □□ backoff(`solar_client.py:107-130`); LLM JSON □□ □ `{...}` □□ fallback; trace_parser□ mangled line skip(printf interleave □□, PR #64); metrics.py □ numerical edge case(division by zero, no baseline)□□ None □□ □ dashboard □ "—" □ □□.

5.2 □ — Key Code Modules

File	Responsibility	Important functions/classes	Input	Output
tools/workload_analyzer.py	workload □□	analyze_workload, save_summary	workloads/*.json	workload_summary.json
tools/llm_advisor.py	LLM □ □/feedback	run_advise, run_feedback, validate	summary + (rules)	recommendation.json, feedback_rules.md
tools/solar_client.py	Solar Pro 3 HTTP client	SolarClient.chat / complete_json, load_env	API □□	LLM text/JSON
tools/algorithm_guard.py	□□ □□ + matrix	validate_recommendation, evaluate_compatibility, PARAM_RANGES, COMPATIBILITY_MATRIX	recommendation.json	guard_decision.json
tools/scheduler_simulator.py	□□□ □□□□□	Simulator, □□□□□ _pick_*	workload + guard	trace_<algo>.jsonl
tools/trace_parser.py	xv6 raw log → JSONL	_PREFIXES, line parser	raw [SCHED]/[SCHEDTEST] log	trace_<algo>.jsonl
tools/metrics.py	□□□ + judgment	compute_metrics, compute_regret, compute_judgment, pick_best_algorithm, evaluate_run	trace + recommendation + baselines	metrics.json
tools/event_detector.py	□□□ □□ □□	detect (4 rule)	trace + metrics	runtime_events.json
tools/correction_proposer.py	□□□ correction rule	propose, _pick_event	runtime_events + recommendation	correction_proposal.json (preview)
tools/correction_guard.py	correction □□□	guard re-run	proposal	correction_guard_decision.json (preview)
tools/trace_explainer.py	LLM □□□ □□	summarize_trace, validate	trace + metrics + rec	trace_explanation.json
tools/schema_compat.py	□ □□□ □□□ □□	normalize_algo, normalize_target_metric, alias □	any JSON	normalized
tools/validate_dashboard_contract.py	dashboard □□ □ □	--preview, --snapshots □□	live-data	exit code
scripts/orchestrator.py	host control plane	--backend xv6/simulator, QEMU □□□	profile	live-data + manifest
scripts/final_demo_check.py	demo □□□□	□□ + □□ smoke	—	log
scripts/multi_profile_demo_check.py	4 profile □□□□	□□ □□	—	log
scripts/export_profile_snapshots.py	snapshot □□	per-profile copy	live-data	snapshots/<profile>/
scripts/correction_preview_smoke.py	preview □□□ offline smoke	□□	—	preview JSON
scripts/analyze_algorithm_winners.py	per-metric 1□ □	comparison parse	metrics	table
xv6-riscv/kernel/proc.c	6□ scheduler □□	scheduler, pick_*, syscalls	xv6	[SCHED] log
xv6-riscv/user/schedtest.c	□□□ □□□□ user prog	find_workload, algo_mode, fork loop	□□	[SCHEDTEST] log
dashboard_live/src/App.jsx	React □□	loadAll, snapshot □□, polling	live-data	UI

6. Scheduling Algorithms and Evaluation Logic

6.1 □□□□□□ □□□ □□□ □□□□

Algorithm	□□	□□□ □□□□	□□
RR (Round Robin)	preemptive, equal time-slice	interactive □□, □□□	`proc.c` `_pick_rr` □□, `simulator._pick_rr`
FCFS	non-preemptive, arrival □	□□ □□□ burst□ / batch	`proc.c:652-682`, `simulator._pick_fcfs`
PRIORITY + Aging	priority □□, 20-tick aging	□□ jobs □□ □□	`proc.c:..-731`, `simulator._pick_priority`
MLFQ	3-queue + quantum [2,4,8] + aging promotion	□□ interactive + □ CPU □□	`proc.c:741-786`, `simulator._pick_mlfq`
SJF (predicted)	non-preemptive, exponential averaging	□□ burst □□ □□ □□□	`proc.c:809-843`
SRTF (predicted)	preemptive SJF	□□ burst□ □□ □□	`proc.c:853-889`

6.2 □□□ □□ (□□□)

```
response_time   = first_run_time - arrival_time
turnaround_time = finish_time - arrival_time
waiting_time    = turnaround_time - total_cpu_burst_time
throughput      = completed_process_count / total_execution_time
```

6.3 □□ □□ □□ (□□ □□)

- `tools/metrics.py:457-467` `compute_judgment`:
  - `starvation_occurred=True` → □□ **FAIL**.
  - □□□ □□□ `regret_score ≤ 0.10 → SUCCESS`, `≤ 0.30 → NEAR-SUCCESS`, else **FAIL**.
- `tools/metrics.py:414-454` `compute_regret`:
  - lower-is-better metric: `regret = (llm − best) / best`.
  - higher-is-better metric(`throughput`): `regret = (best − llm) / best`.
  - □□□ 0□□ clamp.
- baseline□ □□□ PR `d546c20` □□ □□ RR baseline □□. xv6□ □□ trace □□□ PR #15 □ □□□ absolute floor □ □□.

6.4 □□ metrics □□ (`outputs/demo/metrics.json` □□)

Algorithm	avg_response_time	avg_waiting_time	avg_turnaround	throughput	preemption	judgment
MLFQ (LLM □□)	**1.8**	14.8	24.8	0.089	9	**SUCCESS** (regret 0.07)
RR	3.2	20.0	30.0	0.10	10	NEAR-SUCCESS
Priority	8.4	10.8	21.2	0.104	2	NEAR-SUCCESS
FCFS	18.4	18.4	28.8	0.096	0	FAIL
SJF	12.4	12.4	22.8	0.096	0	NEAR-SUCCESS
SRTF	4.8	7.0	17.4	0.104	3	SUCCESS

LLM□ `target_metric=avg_response_time` □□ MLFQ □□ → □□ □□ response_time □□□□□ MLFQ, regret 0.07 < 0.10 → SUCCESS.

---

7. Dashboard and Visualization

7.1 UI □□ (`dashboard_live`)

- 3-column layout (`App.jsx:216-253`).
- **Left:** DemoGuide → LLMRecommendation → AlgorithmGuard → RecommendationEvidence → CounterfactualMetricView → RuntimeCorrectionPreview → EvaluationResult → LLMExplanation.
- **Center:** MainGantt → ProcessState → TraceStack.
- **Right:** ProcessLanes → WorkloadSummary → AlgorithmComparison → MetricVisualization.
- Header: backend badge, snapshot selector, manifest version, last-updated □□, live mode toggle.

7.2 □□□ □□

- `dashboard_live/public/live-data/` (orchestrator□ publish).
- □□ 1 □ □□(`POLL_INTERVAL_MS = 1000`), `manifest.json` □ `version:updated_at` □□ □□ □□ reload.
- snapshot □□: `live-data/snapshots/<profile>/` □ base □ □□, polling □□□□ (□□□□□).

7.3 □□□ □□□ □□ □□

□□	□□□□	□□ □□	□□
Gantt chart	`MainGantt.jsx` + `ProcessLanes.jsx`	DONE	tick □ + algo □ □, PR #4 (`Polish React dashboard`)
Process state	`ProcessState.jsx`	DONE	RUNNING/RUNNABLE/SLEEP/ZOMBIE □
Ready queue (MLFQ)	`ProcessLanes.jsx` queue level □ □□	PARTIAL	□□ “queue □□”□ lanes □□□ □□
Trace event stack	`TraceStack.jsx`	DONE	scroll □□□ □□□ □□□
Workload summary	`WorkloadSummary.jsx`	DONE	analyzer □□ □□□
LLM Recommendation	`LLMRecommendation.jsx`	DONE	algorithm + params + reason + target_metric
Algorithm Guard	`AlgorithmGuard.jsx`	DONE	accept/reject + □□
Recommendation Evidence	`RecommendationEvidence.jsx` (PR #32)	DONE	LLM reason + workload traits + guard score + provenance □ □ □
Metric comparison table	`AlgorithmComparison.jsx`	DONE	□□□□×metric □
Metric visualization	`MetricVisualization.jsx`	DONE	bar/glow + metric selector
Counterfactual metric view	`CounterfactualMetricView.jsx` (PR #43, #44)	DONE	“target metric□ □□□ □□ 1□?”
Evaluation result	`EvaluationResult.jsx`	DONE	judgment(SUCCESS/NEAR-SUCCESS/FAIL) + regret + direction-aware metric □□
LLM Explanation	`LLMExplanation.jsx`	DONE	trace_explanation.json □□
Runtime correction preview	`RuntimeCorrectionPreview.jsx` (PR #54, #56)	DONE (preview only)	□□□ □□□ □□ □□
Demo guide	`DemoGuide.jsx` (PR #46, #47)	DONE	5-step click-to-flash
Live mode	`App.jsx` polling	DONE	snapshot □□□ □□□□

7.4 `dashboard_live` vs `dashboard_test` vs `dashboard/`

□□	`dashboard_live`	`dashboard_test`	`dashboard/`
□□	□□□, □□□□	UI lab, fixture	legacy Streamlit
□□□	`live-data/` □□	`src/data/` □□ fixture	local JSON
□□□□ □	17	16 (Hero/UITestControls □□, evidence/counterfactual/correction/demoguide □□)	□□ Python □□
□□ □□	□□□ □□	UI □□ □	(□□ □ □)

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8. Development Process Summary

8.1 □ □□ (git log + PR □□□ □□)

1. **Phase 1 — □□□ prototype.** workload_analyzer / scheduler_simulator / xv6 RR-baseline □□ □□. workload JSON □□ □□(PR series `b589481..63c7207`).
2. **Phase 2 — LLM □□.** Solar Pro 3 client (`tools/solar_client.py`) □□ → `llm_advisor` → `algorithm_guard` □□. metrics + evaluator □□ □□.
3. **Phase 3 — xv6 scheduler □□.** `setscheduler/getscheduler` syscall, FCFS/PRIORITY/MLFQ □□, □□□□ SJF/SRTF + burst predictor (`3031767 Add prediction-based SJF and SRTF`).
4. **Phase 4 — Dashboard □□.** Streamlit dashboard □□ React/Vite□ □□ (`c88bba6 Add React/Vite observability dashboard`). dashboard □ test/live □□ split (`c95b264 feat: split dashboard into test/live apps`).
5. **Phase 5 — Orchestrator refactor.** “□□□ control plane” □□ (`f6911dd feat(orchestrator): host-side control plane with simulator + xv6 backends`). xv6□ “□□ □□□□ □□□”□ □□ □□, simulator□ dev/fallback.
6. **Phase 6 — Demo readiness.** trace explainer, demo runbook, presenter script, defense notes, RC report, snapshot exporter (4 profile), Recommendation Evidence / Counterfactual / DemoGuide / RuntimeCorrectionPreview □□, CI(QEMU □□ lightweight), validator `--preview` / `--snapshots` □□. PR #20 ~ #69 □ □□□ □ □□.
7. **Phase 7 — RC freeze + revert.** PR #67/#68/#69 (step layout □□□□□□)□ RC freeze □□□ □□□ PR #70□□ revert(`9af5cfb`).
8. **Phase 8 (□□ branch `feat/upstage-runtime-strict`):** Upstage Solar Pro 3 strict □□□ □□ — `2d67299 feat(runtime): strict Upstage Solar Pro 3 backend; opt-in demo fallback` .

8.2 □ — Development Timeline (□□ □□□)

Date / Commit	Change	Related files	Impact	Status
`0b1903d`	root `.gitignore` for credentials	`.gitignore`	API □ □□ □□	DONE
PR #3 / `4742558..6c899e3`	xv6 scheduler harness	`kernel/proc.c`, `user/schedtest.c`	6 □□□□ □□ □□	DONE
`3031767` PR #4	SJF/SRTF + predictor in xv6	`proc.c` (predictor API), `schedtest.c`	predict □□ □□ □□ □□ □□	DONE
`c88bba6`	React/Vite dashboard + host tool pipeline	`dashboard_live/`, `tools/`	Streamlit □□, □□ GUI	DONE
PR #6/#7 / `c95b264` `42840d1`	dashboard split test/live	`dashboard_live`, `dashboard_test`	UI iteration vs □□□□ □□	DONE
`995bc53`	Trace Explainer □□	`tools/trace_explainer.py`	After-running LLM □□	DONE
PR #9 `fed68dc`	live judgment□ selected metric□□ □□□	dashboard	metric □□□ □□ □□ □□	DONE
`f6911dd`	Orchestrator □□ (control plane)	`scripts/orchestrator.py`	xv6/simulator □□□ □□□, schema adapter, dashboard backend □□	DONE
`c8e95e5`	schedtest seed/profile + richer traces	`user/schedtest.c`, `kernel/proc.c`	□□□ xv6 □□, [SCHEDTEST] □□	DONE
PR #11 `108da45`	metrics: derive xv6 arrival_time from PROC_DEF	`tools/metrics.py`	xv6 trace □□□ ↑	DONE
PR #12 `f43687f`	schedtest: planned arrival □□ fork □□	`user/schedtest.c`	□□□□ □□ □□	DONE
PR #14/#15	starvation rule + regret □□□ absolute floor	`tools/metrics.py`	□□ xv6 trace□ □ □□□ FAIL □□	DONE
PR #16 `b68708e`	validator strict mode + manifest checks	`tools/validate_dashboard_contract.py`	dashboard contract CI □□	DONE
PR #17 `0161b9d`	`final_demo_check.py` + printf interleave □□	`scripts/`, `tools/`	□□□□ □□ □□	DONE
PR #25/#26/#27	presenter script + multi_profile_demo_check + CI	`docs/`, `scripts/`, `.github/workflows/`	demo prep □□ □	DONE
PR #35..#38	4 xv6 profile snapshot □□ + selector	`scripts/export_profile_snapshots.py`, `dashboard_live`	dashboard □ 4 □□□□ □□ □□	DONE
PR #41/#43/#44	□□□□ □□□ audit + Counterfactual view + winners table	`scripts/analyze_algorithm_winners.py`, `dashboard_live/.../CounterfactualMetricView.jsx`	“MLFQ□ □□ □ □” □□ □□	DONE
PR #52..#59	Runtime correction **preview** (proposer/guard/RuntimeCorrectionPreview + smoke + validator `--preview`)	`tools/correction_*`, `dashboard_live`, `scripts/correction_preview_smoke.py`	preview-only loop □□□	PARTIAL (xv6 apply □□)
PR #62/#65	RC YELLOW → GREEN (RR row □□□ □□)	`docs/final_release_candidate_report.md`, `scripts/orchestrator.py`	□□ □□ □□ □□	DONE
PR #67/#68/#69 → PR #70	step layout □□□□□□ □ revert	`dashboard_live`	RC freeze □ □ □ □□□	REVERTED
`2d67299` (□ branch)	Strict Solar Pro 3 backend + opt-in demo fallback	`tools/solar_client.py`, `scripts/orchestrator.py`	API □ □□□ □□ fail	DONE

8.3 □□□ □□ / □□ □□

□□□ □□	□□□
xv6 □□ trace□□ starvation □□□	starvation□ absolute floor □□ (PR #14) + regret □□ floor (PR #15)
printf interleave □ `RUN_BEGIN` □□	orchestrator □□□□ substring □□□□ □□ (PR #64); trace_parser □ □□ □□ skip
LLM□ `[1m]` □□ marker□ dashboard □□□	strict JSON mode + validator
dashboard_live□ fallback □□□□ □□□□ □□	manifest `metadata_source` + □□ (`SIMULATOR FALLBACK` / `FALLBACK`) + □□ □□
MLFQ□ “□□ □□”□□ □□□ □□ □□	analyze_algorithm_winners + CounterfactualMetricView □ metric□ □□□ □□□□ □□□

□□ □□	□□
Runtime correction □ closed-loop xv6 apply	`tools/correction_proposer.py` □□□ □□ preview only
Feedback rule □□ □□□	`tools/llm_advisor.py --mode feedback` □ □□ □□
MLFQ □ □□□□ □□□□ cover □□	4 profile □□ LLM□ MLFQ □□ → □□□ audit(`docs/algorithm_decision_diversity_audit.md`)□□ □□
QEMU □□ □□ CI□ lightweight smoke □	□□ xv6 □□ CI□ □□

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9. Current Limitations and Risks

1. **xv6 integration□ "□□ + optional"□ □□.** `scripts/orchestrator.py --backend xv6` □ □□□ QEMU □□ + `schedtest` □□ + □□ capture□□ □□. □, □ □ □□□□ QEMU□ □□ □□ □□ □□□ simulator fallback□ 4□ xv6 snapshot□ □□□□ □□.
2. **xv6 traces are short (~30–80 events per algorithm).** `schedtest` □ curated profile□ 4-5□ child□ fork. metric □□□□□□ □□□□□ □□□ □□.
3. **LLM API □□□ □□(real).** `tools/solar_client.py` □ □□ `https://api.upstage.ai/v1/chat/completions` □□. □□ □□□ orchestrator □ `--offline-fixture` □ □□ `outputs/demo/recommendation.json` □□ □□.
4. **Trace schema □□□:** simulator(`time` / `algorithm`)□ xv6(`tick` / `algo`) □□□ □□□□ `tools/schema_compat.py` □ `tools/metrics.py:62-77` `normalize_event` □ □□. □ □□□ □□ □□ □□□ □□□ □□ □.
5. **Metrics □□□:** trace□□□ □□□ + xv6□ EXIT □□ □□ □□ □ □ □□. □□ trace□□□ starvation/regret edge case□ PR #14/#15 □ □□.
6. **Dashboard □□ □□:** runtime correction □□□ "preview" □□ □□. □□□ □□□ □□ □□. snapshot selector□ □□□□□ polling □ □.
7. **Demo scenario:** `docs/demo_runbook.md`, `docs/demo_checklist.md`, `docs/presenter_script.md` □ □□ □ □□. README §10 □ "starvation □□□□"□ closed-loop□ □□□□ Future Work□□ □□.
8. **Test coverage:** unit-test □□□□□ □□. □□ `scripts/final_demo_check.py`, `scripts/multi_profile_demo_check.py`, `scripts/correction_preview_smoke.py`, `tools/validate_dashboard_contract.py` □ "end-to-end smoke + □□ □□" □□. CI□ `.github/workflows/` □ lightweight smoke □ (QEMU □□□).
9. **README □□ □□ □□□:** README□ docs□ □□(`make qemu`, `python3 tools/scheduler_simulator.py`, `streamlit run dashboard/dashboard.py`, `cd dashboard_live && npm run dev`)□ □□ □□□ □□. □ `tools/scheduler_simulator.py` □ □□ □□ □□□□ □□□□/□□ □□ □□ → orchestrator □□□ □□ □□.
10. **API key / secret □□:** `.gitignore` □□ + `.env.example` placeholder + `SolarClient` □ □ □□□ raise. **□□□□ □ □□ □□□.**
11. **□□ □□□□ □□□□ □□ □ □□ □□:**
    - "LLM□ □□ MLFQ□ □□□□ LLM□ □□□ □□?" → CounterfactualMetricView + algorithm_decision_diversity_audit □ metric□ □□□□ □□□□ □□ □□ □□□.
    - "runtime correction□ closed-loop □□?" → □□□□ preview only. README/docs/§12.1□ □□.
    - "xv6 schedtest□ □□ OS workload□ □□□ □□?" → □□□ / □□□ / □□□□ □□□□ □□□ □□. `docs/presentation_defense_notes.md` □ □□ □□.

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10. Next Action Plan

□ — Next Tasks

Priority	Task	Why it matters	Related files	Owner role (□□)	Expected outpu
P0	□□□ dashboard □ □ □ dry-run + screenshot □□ (4 profile □□ swipe)	□□□ □□□ □□ □□	`dashboard_live/public/live-data/snapshots/*`, `docs/presenter_script.md`	Frontend/Demo lead	`docs/final_dem` □□ + screenshot
P0	`.env` □□ □□□□ □□□ □□□ □ □ □ □□	□□ □ API □ □□□ □□	`tools/solar_client.py`, `.env.example`, README §3	Backend/LLM	README □□ dem shoot □□
P0	4 xv6 snapshot □□□ □□□ (`tools/validate_dashboard_contract.py --snapshots`)	dashboard “FALLBACK □□” □ □□□ □□ □ □	`dashboard_live/public/live-data/`, validator	DevOps	exit 0 □□
P0	RC freeze □ □□ □□ branch (`feat/upstage-runtime-strict`) □□ □□ □□	strict □□ vs RC GREEN main □ □□ □ □□	`scripts/orchestrator.py`, `tools/solar_client.py`	Tech lead	□□ □□ □□ □ □□ □□
P1	Runtime correction□ xv6 apply step □□ (`setscheduler` □□□ + `CORRECTION_APPLIED` event emit)	closed-loop □ □ □ □□□□ □□□	`kernel/proc.c`, `user/schedtest.c`, `scripts/orchestrator.py`	xv6 lead	xv6 apply □□ + □
P1	Feedback loop □□ □□□ (orchestrator□ FAIL □ `--mode feedback` □□ □ next-run prompt□ □□ append)	LLM□ “□□” □□ □□ □□	`scripts/orchestrator.py`, `tools/llm_advisor.py`	LLM lead	□□ □□ + dashboa
P1	MLFQ □ □□□ □□□ □□ workload (□: □□ batch□ → SJF, single CPU bound → FCFS) □□ + LLM □□ □□ □□	□□ □□□ “□□”	`workloads/`, `docs/algorithm_decision_diversity_audit.md`	LLM/eval	□□ workload + w
P1	`tools/algorithm_guard.py` □ SJF/SRTF predictor □□□ □□ □□□	edge case □□	`tools/algorithm_guard.py`	guard owner	□ check + □□□
P2	unit-test □□ (`tools/metrics.py` judgment, `tools/correction_proposer.py` □ □□□)	regression □□	`tests/` □□	dev	pytest □□
P2	Streamlit `dashboard/dashboard.py` □□ □□ archive □□	repo □□□□ □□	`dashboard/`	repo □□	□□ PR
P2	Live mode polling□ SSE/WebSocket□ □□	UX □□	`dashboard_live/src/data/`, `scripts/`	frontend	streaming endpo
P2	xv6 schedtest □ □ □ workload (10-20 child)	metric scale □□□	`xv6-riscv/user/schedtest.c`	xv6	□□ profile

P0 □□□ □□ “demo□ □□□□ □□ □□ □” □.

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11. Final Demo Readiness

□□	□□ (0~5)	□□
End-to-end pipeline readiness	**5**	simulator □□□□ □□ □□ (`scripts/orchestrator.py --backend simulator`). xv6 □□□□ QEMU □□□ □□. 4 profile snapshot □ □□□ QEMU □□ □□□□□ □□ □□. dashboard contract validator + CI smoke.
xv6 integration readiness	**4**	6 □□□□ □□ □□ □□, syscall □□, `schedtest` □□□ □□□□, orchestrator□ QEMU build/boot/capture. □ closed-loop runtime correction □ apply step □□□.
LLM advisor readiness	**4**	Solar Pro 3 strict □□ + JSON □□ + validator + feedback □□. □□ feedback □□□□ □□. 4 profile □□ MLFQ □ □□□□ □□ □□□ □□.
Metrics / evaluation readiness	**5**	response/turnaround/waiting/throughput/starvation/regret/judgment □□. starvation □□□ floor + □□ RR baseline + best_algorithm winners.
Dashboard readiness	**5**	17 □□□□, snapshot selector, backend □□, DemoGuide, RuntimeCorrectionPreview, Recommendation Evidence, Counterfactual view. `dashboard_live` □ □□□ □□.
Documentation readiness	**5**	README 29 KB + `docs/*.md` 27 □. presenter_script / demo_runbook / demo_checklist / defense_notes / RC report / multiple audits.

□□: □□ □□□ **GREEN** (PR #65). xv6+QEMU □ □□ □□□□□ snapshot 4□□□ □ □□ □□. closed-loop correction□ “Future Work” □□ □□ □□ □□□□ □□ □□□ □□.

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12. Conclusion

12.1 □□ 3

1. **“LLM□ □□□, xv6□ □□”□ □□ □□□ □□ □□□□ □□□.** `algorithm_guard` + `correction_guard` □ LLM □□□ □□□□ □□□□ □□ □ □□□ □□□□. `correction_proposer/guard` □□□□ `preview_only=true / applied=false` . LLM □□ □□□ `tools/solar_client.py` □□□ □□.
2. **□□ □ □□□□□□ JSON/JSONL □□□□ □□ □□□□□ □□□□ □□·□□ □□.** `tools/schema_compat.py` □ simulator/xv6 □ □□□□ trace schema □□□ □□□□, `tools/validate_dashboard_contract.py` □ dashboard □□□ CI□ □□.
3. **xv6 6□ □□□□□ □□□ □□□ □□□□ □□, `schedtest` + orchestrator □ □□□□□ □□□□.** □□ □□□□□ □□□ □□□ “□□ OS □□ trace” □ metric□ □□□□. 4 profile snapshot □□ □□ □□□□ □□ □□□□□ □□ □□ □□.

12.2 □□ □□□ □□ 3

1. **Closed-loop runtime correction□ preview-only.** README□ “starvation □□□□” □ □□□□□(LLM □□ → xv6 □□ → □□ □□)□ □□ □□□□ □□. preview UI□ □□ “□□ □”□□ □□ □□.
2. **LLM □□ □□□ □□.** 4 profile □□ MLFQ□ □□ → “□ LLM□ □□ MLFQ □ □□□” □□□ □□ □□(Counterfactual + winners)□ □□ □□□ □□ □□□ □ workload(□: □□ batch) □ □□ SJF/FCFS □□□ □□□□ □□□ □ □□.
3. **□□□ □□□ □□.** □□ □□□□ □□ □□□□ smoke □□□□ + contract validator + CI lightweight □. metric □□ / judgment / correction □□ □□□ PR □□□ □□□□ □□ □.

12.3 □□ □ □□□ □□□ □ □ 3

1. **□□ branch `feat/upstage-runtime-strict` □□ □□ □□ + □□ 4 snapshot □ validator □□ □□□** (□ P0).
2. **demo dry-run 1□ + screenshot □□ □□** (□ P0).
3. **README/demo_runbook □ “API □ □□ □ fallback □□ □□□ □” □ □□ □□** — `--offline-fixture` □□□□ `FALLBACK` □□ □□□ □□□□ □□□ □ □□□ (□ P0).

Appendix — □□ □□

- **□□□ □□ □□ □:** □ 35 □ (Python 18, JSX 17, xv6 C 2, JSON/JSONL/MD □□□ sampling □□).
- **□ □□□ (Python+JSX+xv6 C □□):** Python tools+scripts ≈ 6.3 k LOC, dashboard React ≈ 17 components, xv6 □□ ≈ 1.4 k LOC.
- **Git □□ □□:** □□ 100 commits + main□ PR □□□.
- **□□ □□ □□:** □□ QEMU □□ □□ (□ □□ □□□□ QEMU □□□). □□ `outputs/xv6_raw_*_seed42.log` , `outputs/check_xv6_scheduler_*.log` , `outputs/build_*.log` □ □□□ □□ □□ □□□ □□□.
- **□ □□□□ API □ / `.env` □□ / □□ secret □□ □□□□ □□.**